Worksheet 6-10 A
Integrals involving Long Division and Completing the Square
AP Calculus Name_____

Find the indefinite integral:

1.) $\int \frac{1}{x^2+6x+10}\,dx$

2.) $\int \frac{5x^2-31x-20}{5x+4}\,dx$

3.) $\int \frac{5}{\sqrt{-x^2-4x-3}}\,dx$

4.) $\int \frac{7}{x^2-4x+5}\,dx$

5.) $\int \frac{1}{3\sqrt{-x^2-10x-24}}\,dx$

6.) $\int \frac{8}{4x^2+24x+39}\,dx$

7.) $\int \frac{4x^2}{x-2}\,dx$

8.) $\int \frac{42x^2-10x-90}{7x+10}\,dx$

9.) $\int \frac{16}{x^2-6x+25}\,dx$

(A) $16\ln|x^2-16x+25|+C$

(B) $\arctan\left(\frac{x-3}{4}\right)+C$

(C) $16\arctan(x-3)+C$

(D) $4\arctan\left(\frac{x-3}{4}\right)+C$

10.) $\int \frac{12}{\sqrt{-x^2-2x+3}}\,dx$

(A) $12\arcsin\left(\frac{x+1}{2}\right)+C$

(B) $12\arcsin\left(\frac{x-1}{2}\right)+C$

(C) $6\arcsin\left(\frac{x-1}{2}\right)+C$

(D) $6\arcsin\left(\frac{x+1}{2}\right)+C$